

EE 524: Digital MOS LSI Circuits

Course Code : EE 524

Course Name : Digital MOS LSI Circuits

L-T-P-C : 3-0-0-3.

Pre-requisite : IC161 - Applied Electronics or Equivalent, EE311-Device Electronics for Integrated Circuits or Equivalent

Intended for : BTech Electrical Engineering (EE), M. Tech., MS & PhD.

Distribution : Elective for Third and Final year B. Tech (EE), MS, M. Tech. & PhD

Approval: 14th Senate

Course Contents

- **Introduction:** An overview of IC development and trends. A review of basic properties of MOS transistors and device physics relevant for digital logic design. CMOS process technology, layout and design rules. [5 Lectures]
- **CMOS Inverter:** Static CMOS inverter, static behavior switching threshold, noise margin, robustness. Dynamic behavior capacitance computation, propagation delay- first order analysis, power, energy and energy delay. [3 Lectures]
- **Combination logic gates in CMOS:** Static CMOS design complementary CMOS, Ratioed Logic, Pass transistor logic- dynamic CMOS Design dynamic logic principles, speed and power dissipation, issues in dynamic design, cascading dynamic gates, designing logic for reduced supply voltage, simulation and layout techniques for complex gates. [5 Lectures]
- **Sequential logic circuits in CMOS:** Timing metrics for sequential circuits, memory element classifications, static latches and Registers, dynamic latches and Registers, Alternative register styles pulse and sense amplifier based registers, pipelining, Non Bi stable sequential Circuits, Choosing clock strategy. [5 Lectures]
- **Design Criteria:** Introduction Custom, semi custom and structured array design approaches, cell based Design methodology, Array based implementations pre-diffused pre-wired arrays, characterizing logic and sequential cells. [5 Lectures]
- **Interconnect Gilbert Strang:** Coping with interconnect , capacitiveparasitics cross talk, resistive parasitics Ohmic voltage drop- electromigration-RC delay, inductive parasitics voltage drop transmission line effects, advanced interconnect techniques reduced Swing Circuits Current mode transmission Techniques. [4 Lectures]
- **Timing issues:** Timing classification of digital systems, synchronous interconnect, Synchronous timing basics, source of skew and jitter, clock distribution techniques and latch based clocking, self timed circuit design- clock synthesis and synchronization using a phase Locked loop. [5 Lectures]
- **Design verification:** Datapaths in digital Processor architectures, the adder, multiplier, shifter, power and speed tradeoffs in datapath structures -memory architecture and buliding blocks, memory core, peripheral cicuit, reliability and yield, power dissipation. [5 Lectures]
- **Design for Testability:** issues in design for testability, ad hoc testing, scan based testing, boundary scan design, Built in self Test, test pattern generation, fault models. [5 Lectures]

Text Books:

1. Neil H.E. Weste and Harris D M, **CMOS VLSI Design: A circuit and Systems Perspective**, Fourth Edition, Addison Wesley, 2011.

References:

1. Jan M. Rabaye, **Digital Integrated Circuits: A Design Perspective**, 2nd Edition, Prentice-Hall, 2003.
2. Ken Martin, **Digital Integrated Circuit Design**, Oxford University Press, 2000.
3. Sung-Mo Kang, Yusuf Leblebici, **Digital Integrated Circuits: Analysis and Design**, McGraw-Hill, 2002.
4. A. Chandrakasan, W. Bowhill, F. Fox, **Design of High Performance Microprocessor Circuits**, IEEE Press, 2000.
5. John P. Uyemura, Thomson, **Chip Design for Submicron VLSI: CMOS Layout and Simulation**, 2005.